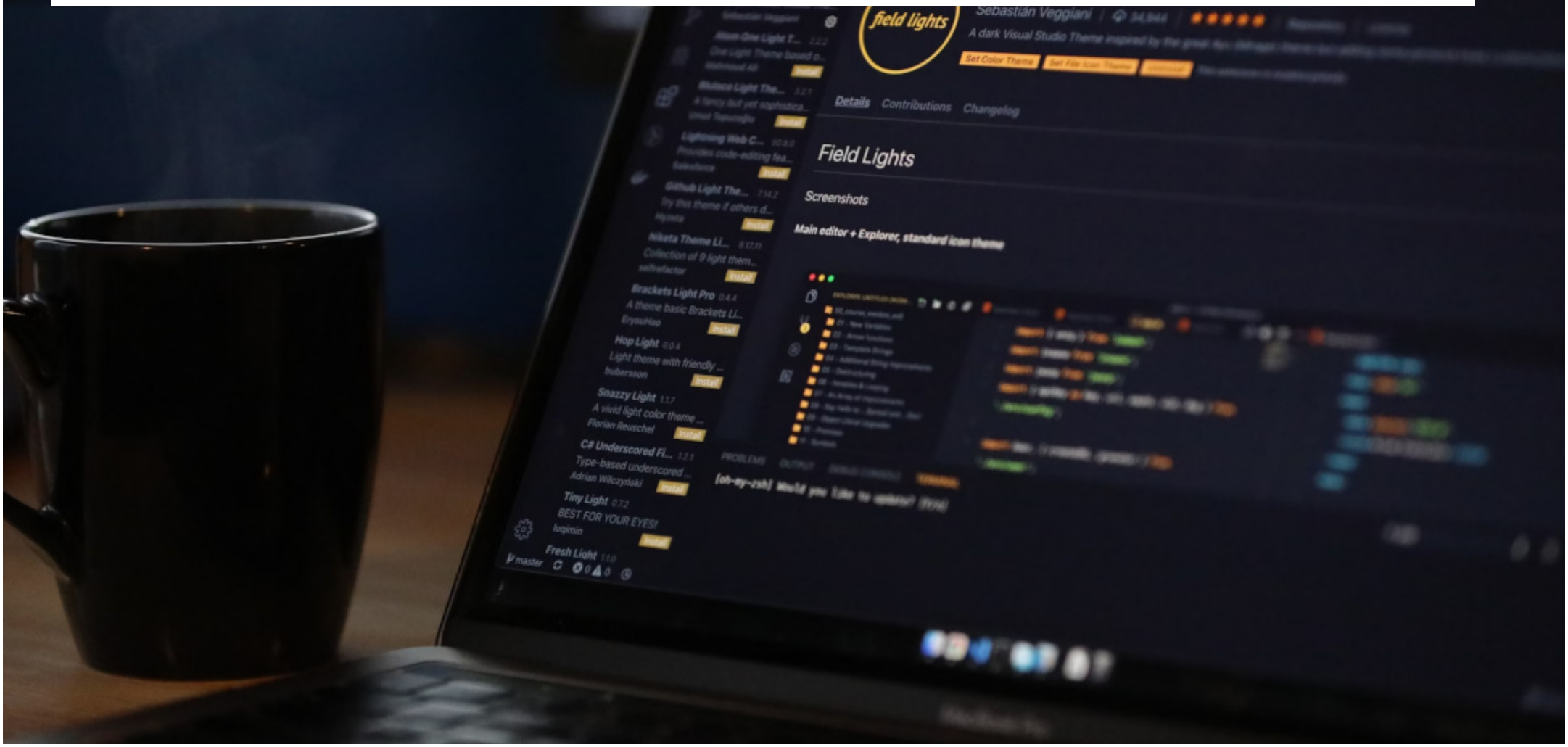
sp_blitz SQL Server Risk Assessment

SQLDBA

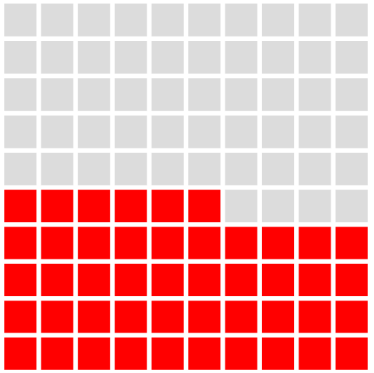
A SQL report based on data from the First Responder Kit, https://www.brentozar.com/responder/

We use data from using the following parameters for the stored procedure: EXEC [dbo].[sp_Blitz] @CheckUserDatabaseObjects = 1 , @CheckProcedureCache = 1 , @OutputProcedureCache = 0 , @CheckServerInfo = 1, @BringThePain = 1;



1. SQL health score*

46%



40
High Priority

40
Low Priority

272
Passed Checks

If you are unsure about who owns the risk on this page, pass this on to your manager and request that they acknowledge the potential risks. This will limit the fallout of resume generating events.

The following potential operation risks were evaluated:

Failed items listed below could potentially void your SQL support SLA

Risk Score	Category	Outcome	
9	Configuration	Some SQL servers are incorrectly licensed. There is a risk of financial liability. Take urgent action.	!
12	Monitoring/Alerting	Configure additional SQL alerts. This will enable early issue detection.	!
12	Monitoring/Alerting	Configure corruption alerts. Corruption leads to data loss. Immediate notification of corruption is required. Take urgent action.	!
25	Reliability	Some databases have no backups. Get SQL backups configured now. Take immediate action.	!
15	Reliability	1 databases have no recent LOG Backups	!
9	Reliability	There are data consistency risks. These could lead to data loss. Take immediate action.	!

User Databases: **5** SQL Instances: **1**

* The SQL health score is a weighted score based on the complexity and impact of failed checks or items that require attention.

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2. SQLDBA triage Actions

This report contains a lot of information which we realise can take some time to digest, evaluate and develop an action plan.
To give a head start we have highlighted some items which we deem as essential and relatively easy to implement. These items are non-impacting on the schema/structure, so the application vendor or DevOps team don't have to go through a separate approval process.
Within your environment these are the DBA actions that should be prioritised:

Rank	Action	Likelihood (1-5)	Consquence (1-5)	Risk
1	Some SQL servers are incorrectly licensed. There is a risk of financial liability. Take urgent action. • Contact licensing partner to validate current Microsoft licensing position. • Review and correct the licensing risk and liability to your business. Engage your DBA to action this. • Replace license wit appropriate licences.	3	3	**9**
2	Configure additional SQL alerts. This will enable early issue detection. • Enable additional Microsoft SQL automated email alert notifications. Engage your DBA to action this	3	4	**12**
3	Configure corruption alerts. Corruption leads to data loss. Immediate notification of corruption is required. Take urgent action. • Enable Microsoft SQL automated email alert notifications. Engage your DBA to action this.	3	4	**12**
4	Some databases have no backups. Get SQL backups configured now. Take immediate action. • Configure automated backups to a local disk, or file share. Engage your DBA to action this. • your DBA recommends weekly FULL backups, daily DIFFERENTIAL and 15-minute LOG backups	5	5	**25**
5	Some databases do not have LOG backups. This can lead to data loss, blow out disk space or cause an outage. Take immediate action. • Ensure SQL LOG backups are taken every 15-minutes. Engage your DBA to action this. • Backup to a local disk, or file share	5	3	**15**
6	There are data consistency risks. These could lead to data loss. Take immediate action. • Ensure consistency checks are done daily. Engage your DBA to action this. • These can be automated	3	3	**9**

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3. SQL servers in scope

The following servers were in scope for this report.

SQL instance	Edition	Version	Support Ending	Missed patches	Current licensing USD*	Databases
MSI	Developer	16.0.4185.3	11/01/2033	11	$0	5

* Annual current licensing estimated on a 3 year Open value agreement with SA. Based on Edition and CPU count

4. SQL server usage overview

The following usage summary highlights usage patterns across the SQL estate and can assist in evaluating servers for consolidation.

SQL instance	Edition	CPUs	Sockets	CPU Speed	Memory	DBs on server
MSI	Developer	22	2	3.07GHz	31GB	5

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5. Recommended actions

Some of these items might not be covered under any existing SQL support agreement. **Issues are generally ranked highest to lowest priority in each category**. Category priority ranking can be determined by the business based on the CIA (Confidentiality, Integrity, and Availability) triad of information security.
The CIA triad is an information security benchmark model used to evaluate the information security of an organization. Best to attend to these sooner rather than later.

Key	Description
✓	Check passed.
🔍	May require further investigation
!	Check failed.

Key	Impact
	Low impact
	Medium impact
	High impact

Key	Complexity
○	Low complexity
◐	Medium complexity
●	High complexity

Reliability - Integrity & Availability

Check	Summary	Details	Impact	Complexity	Issues	Effort required
!	There are data consistency risks. These could lead to data loss. Take immediate action.	Database integrity needs to be checked daily. The data integrity on these servers cannot be validated if these checks aren't run		●	1	2
!	Some databases have no backups. Get SQL backups configured now. Take immediate action.	Regular backups of databases and transactions are critical for achieving RPO and RTP targets.		◐	1	5
!	1 backups done to a disk hosting databases	Backing up to a drive where databases reside becomes a problem when a failure impacts that drive.		◐	1	
🔍	1 Logged High Impact SQL Admin Tasks	These are normally high impact DBA DBCC tasks. This does not include CHECKDB and other usually benign DBCC events.		◐	1	2
!	1 databases have no recent LOG Backups	Without Log backups databases can consume disk space and database restores are compromised.		○	1	2
!	Backup Compression is NOT enabled on all server(s). It is easy to configure and improves backup and restore times	Compressing backups makes backups and restores perform much better.		○	1	12
!	1 Errors found in the SQL Trace Log	The Errors Logged by in the SQL server Logs needs to be monitored.		◐	1	3
!	Remote Admin connections are enabled on all server (s)	The Dedicated Admin Connection (DAC) allows Admins to connect to the SQL server during certain problem scenarios		○	1	6
🔍	1 server(s) restarted in last 24 hours	Checking server restart times improves our ability to interpret monitoring data		○	1	

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Performance - Availability

Check	Summary	Details	Impact	Complexity	Issues	Effort required
!	Poison Wait Detected	Some wait types will often indicates killer performance problems. We check for records of these on the SQL server since last restart.			1	
	Instance Stacking	Instance Stacking. Multiple instances requires special configuration considerations.			1	
	Some Waits are concerning	Waits indicate whether the server seems to be relatively idle, or the server was restarted recently.			1	
!	4 RID or Key Lookups	A top resource-intensive queries contains RID or Key Lookups. Requires covering indexes.			4	
!	Plan Cache cleared recently	When someone runs DBCC FREEPROCCACHE, restarts the SQL Server, or the server is under horrific memory pressure the plan cache gets cleared			1	
!	Possible parallelism threshold configuration issues	Parallelism threshold is set by default to 5. Changing this may reduce CXPACKET waits.			1	4
	Found 1 Scalar UDFs	A top resource-intensive queries is using a user-defined scalar function that may inhibit parallelism.			1	
!	Some SQL server(s) do not have query plan memory optimization enabled. There are no reasons why this should be disabled.	Query plan memory utilization can be optimized for SQL server. The default setting keeps this disabled.			1	
!	Instant File Initialization Disabled on 4 server(s)	The Perform Volume Maintenance Tasks permission allows the SQL server to complete database file growth operation faster			4	
!	1 Implicit Conversions impacting SQL queries	Implicit conversion affects cardinality estimation, which is crucial in query execution.			1	
!	1 Indexes Disabled	Disabled index are not actually helping performance and should either be enabled or removed.			1	
	4 Cursors found	Some resource-intensive queries use cursors. This may impact performance.			4	
	1 Implicit Conversions in top queries	One of the top resource-intensive queries is comparing two fields that are not the same datatype.			1	
	1 Agent Jobs Starting Simultaneously	When multiple SQL Server Agent jobs are configured to start simultaneously it can have unexpected impacts on server resources like CPU and disk I/O			1	
!	Asynchronous Stats Updates enabled on 1 databases	This may impact performance on queries that trigger the statistic updates.			1	
	1 Single-Use Plans in Procedure Cache	This may be a good use case for enabling the SQL Server 2008 memory optimization and parameterization features			1	6
	1 Stored Procedure WITH RECOMPILE	This may cause increased CPU usage due to constant recompiles of the code.			1	

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Configuration - Availability & Integrity & Confidentiality

Check	Summary	Details	Impact	Complexity	Issues	Effort required
!	1 server(s) are running bad default settings for parallelism	Cost threshold for parallelism. Specifies the threshold at which SQL Server creates and runs parallel plans for queries			1	
	Other SQL services, like SSAS/SSIS/SSRS, are Installed	These services may affect performance			1	
	Default Config Changed on 1 databases	Snapshot Isolation Enabled. Once snapshot isolation is enabled, updated row versions for each transaction are maintained in tempdb			1	
	Default Config Changed on 1 databases	Read Committed Snapshot Isolation. Determines the behavior of the default READ COMMITTED isolation level when snapshot isolation is enabled in a database			1	
	Default Config Changed on 4 server(s)	max degree of parallelism. The degree of parallelism sets the number of processors employed to run a single statement, for each parallel plan execution			4	
!	Some SQL servers are incorrectly licensed. There is a risk of financial liability. Take urgent action.	Running production workloads will incur Licensing fees			4	24
	Default Config Changed on 1 databases	Containment Enabled. A contained database is a database that is isolated from other databases and from the instance of SQL Server that hosts the database			1	
!	1 server(s) are running bad default settings for Backup checksum	Backup checksum default. Provide error-management options (CHECKSUM and NO_CHECKSUM) when using backup applications or utilities that don't natively expose these options			1	
	Some User Databases on C Drive	Putting databases on the C drive runs the risk of crashing the server when it runs out of space.			4	12
	Default Config Changed on 1 databases	Auto Create Stats Incremental Enabled. The statistics created are per partition statistics. When OFF, the statistics tree is dropped and SQL Server recomputes the statistics.			1	
!	1 server(s) are running bad default settings for Agent XPs	Agent XPs. This enables the SQL Server Agent extended stored procedures on this server.			1	
!	1 server(s) are running bad default settings for Database Mail XPs	Database Mail XPs. Used to enable Database Mail on this server.			1	
	TempDB on C Drive	TempDB frequently grows unpredictably, putting your server at risk of running out of C drive space and crashing.			4	12
	Some System Database on C Drive	Putting databases on the C drive runs the risk of crashing the server when it runs out of space.			4	1
	Default Config Changed on 1 databases	Forced Parameterization Enabled. As part of parameterizing a query, SQL Server assigns a data type to the parameters that replace the literal values, depending on the value and size of the literal.			1	
!	1 TraceFlags are configured. Some are missing	These custom switches enable or disable specific behaviour. In most cases these have positive impacts.			1	
!	4 server(s) are running bad default settings for server memory	Max server memory (MB). This should be set and the default should not be usedLimits the size of the SQL Server buffer pool.			4	
!	1 server(s) are running bad default settings for replication size	Max text repl size (B). Specifies the maximum size (in bytes) of text, ntext, varchar(max), nvarchar(max), varbinary(max), xml, and image data that can be added to a replicated column or captured column in a statement.			1	

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	There are 1 Tables in the Master Database.	Unexpected usage of the Master database can cause data growth which may lead to longer recovery times.			1
	Default Config Changed on 4 databases	Service Broker is Enabled for this database.			4
	Default Config Changed on 1 server(s)	contained database authentication. Specifies whether contained databases are allowed in the SQL instance.			1
	Default Config Changed on 1 server(s)	recovery interval (min). This option defines an upper limit on the time recovering a database should take.			1
	Default Config Changed on 1 server(s)	remote admin connections. Used by DBAs to execute diagnostic functions even when the server is locked or running in an abnormal state and not responding to a SQL Server Database Engine connection.			1
	Default Config Changed on 1 server(s)	xp_cmdshell. This option allows system administrators to control whether the xp_cmdshell extended stored procedure can be executed on a system			1

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Monitoring/Alerting - Availability

Check	Summary	Details	Impact	Complexity	Issues	Effort required
!	Configure corruption alerts. Corruption leads to data loss. Immediate notification of corruption is required. Take urgent action.	These three errors can give you notification about early hardware failure			1	6
!	Configure additional SQL alerts. This will enable early issue detection.	These are some very severe SQL Server errors.			1	12
!	1 Agent Jobs Without Failure Emails configured	No email will be sent if it fails.			1	1
!	Failsafe Operator Not present	It is a good idea to enable a default send to location			1	6
!	Some other Alerts are missing or not configured	Not all SQL Server Agent alerts have been configured			1	6
!	SQL Operators are Not Configured	No SQL Server Agent operators (emails) have been configured			1	6
!	1 Alerts do not have Follow Up Configured	Alerts are configured but they either do not notify anyone or else they do not take any action			1	6
	1 Alerts Without Event Descriptions	Having detailed event descriptions in Alert output messages allows for better monitoring.			1	
!	Agent is currently offline	The SQL Agent allows scheduled maintenance and jobs to be performed. This service should always be running.			1	

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Security - Confidentiality

Check	Summary	Details	Impact	Complexity	Issues	Effort required
	1 SQL Jobs are Owned by Users	Disabling these users in AD will cause the job will stop working.			1	0
	4 databases have Owners which is not 'sa'	Databases are owned by other accounts, these accounts might have more permissions than they need.			4	
	SQL Server is running under an NT Service account	SQL Server services can run as NT Service accounts. It is ideal to run as Active Directory accounts.			4	
	SQL Server Agent is running under an NT Service account	SQL Server Agent services can run under NT Service accounts. It is ideal to run as Active Directory accounts.			4	
	4 Sysadmins are present	Sysadmins can do absolutely anything in SQL Server. Modifying security settins amongst others.			4	
	4 Users have Elevated Permissions on a Database	These users can perform tasks beyond just reading and writing data.			4	

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